

Multimodal Disaster Damage Segmentation

RGB + SAR Temporal Fusion for Pixel-Level Damage

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GitHub Repository: <https://github.com/Johnson-Opadere/multimodal-damage-segmentation>

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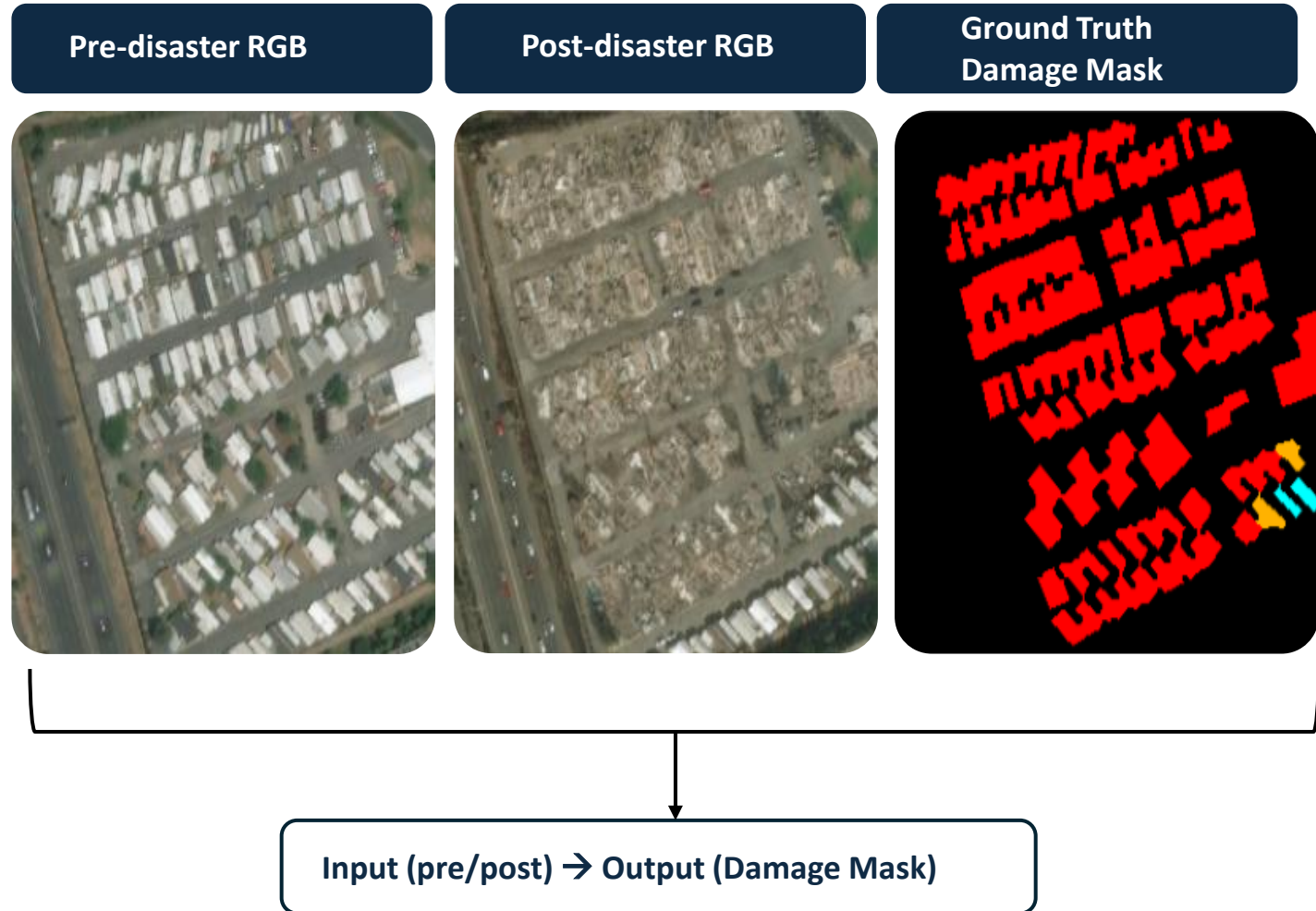
Disaster Damage Segmentation

Goal: pixel-level classification under real-world conditions

What we want to predict

- 0** Background
- 1** Minor Damage
- 2** Major Damage
- 3** Destroyed

Accurate Damage Masks enable faster response and better resource allocation



Why RGB Alone Fails – and Multimodal Fusion Succeeds

Challenge: images are affected by lighting, shadows, smoke, and occlusions, leading to ambiguous imperfections.

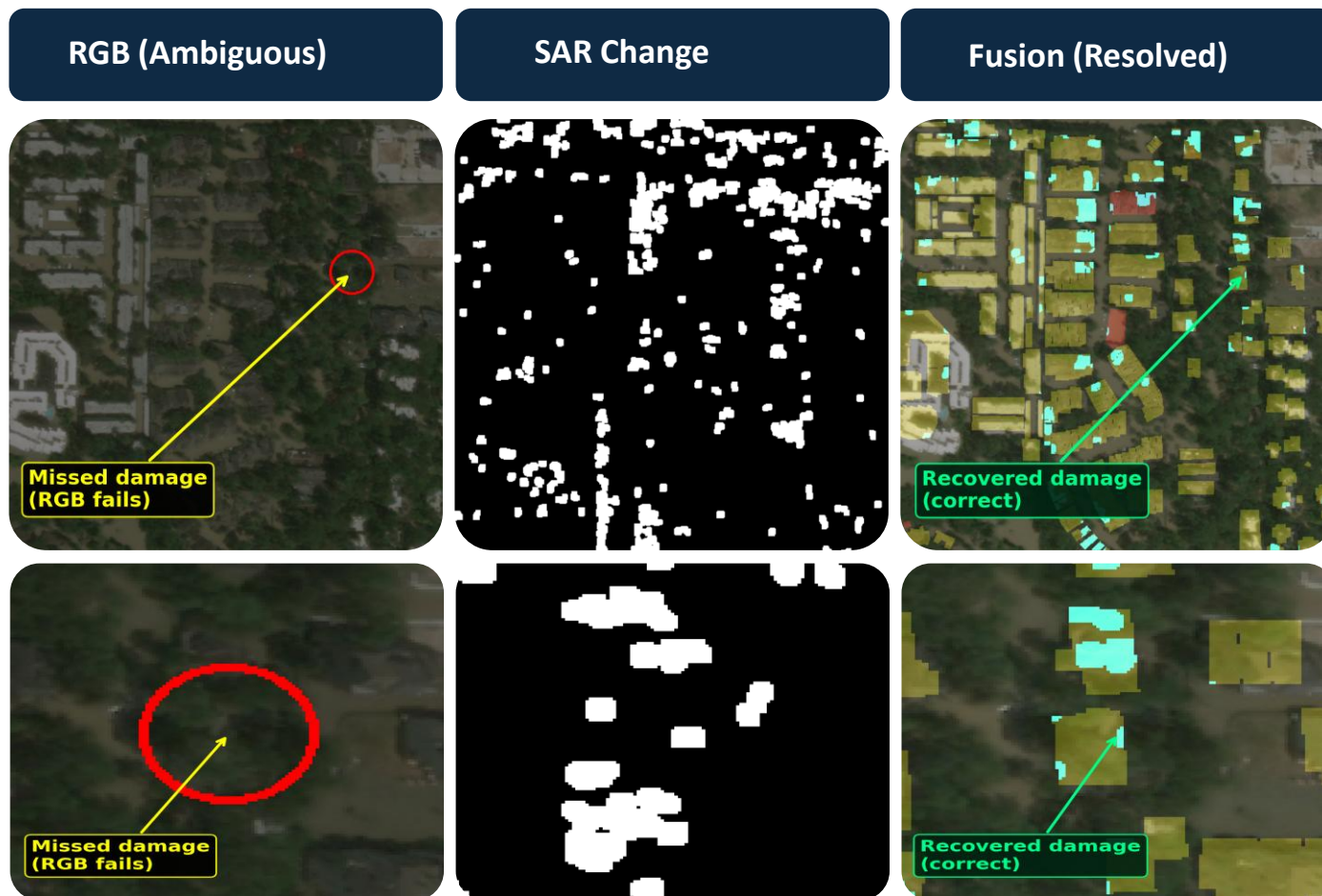
SAR captures structural changes that RGB misses.

Fusion combines both signals recovering accurate damage.

What this example shows

- RGB fails to detect a damaged area (ambiguous appearance)
- SAR provides complimentary structural-change information, while multimodal fusion improves damage detection in ambiguous scenes

Multimodal Fusion resolves ambiguity in complex scenes and improves damage detection reliability



Data Pipeline – from Raw Data to Model Output

Data Sources:

- xView2 RGB (pre + post disaster)
- Sentinel-1 SAR (pre + post disaster)
- Annotations (xView2 WKT Polygons)

Challenges:

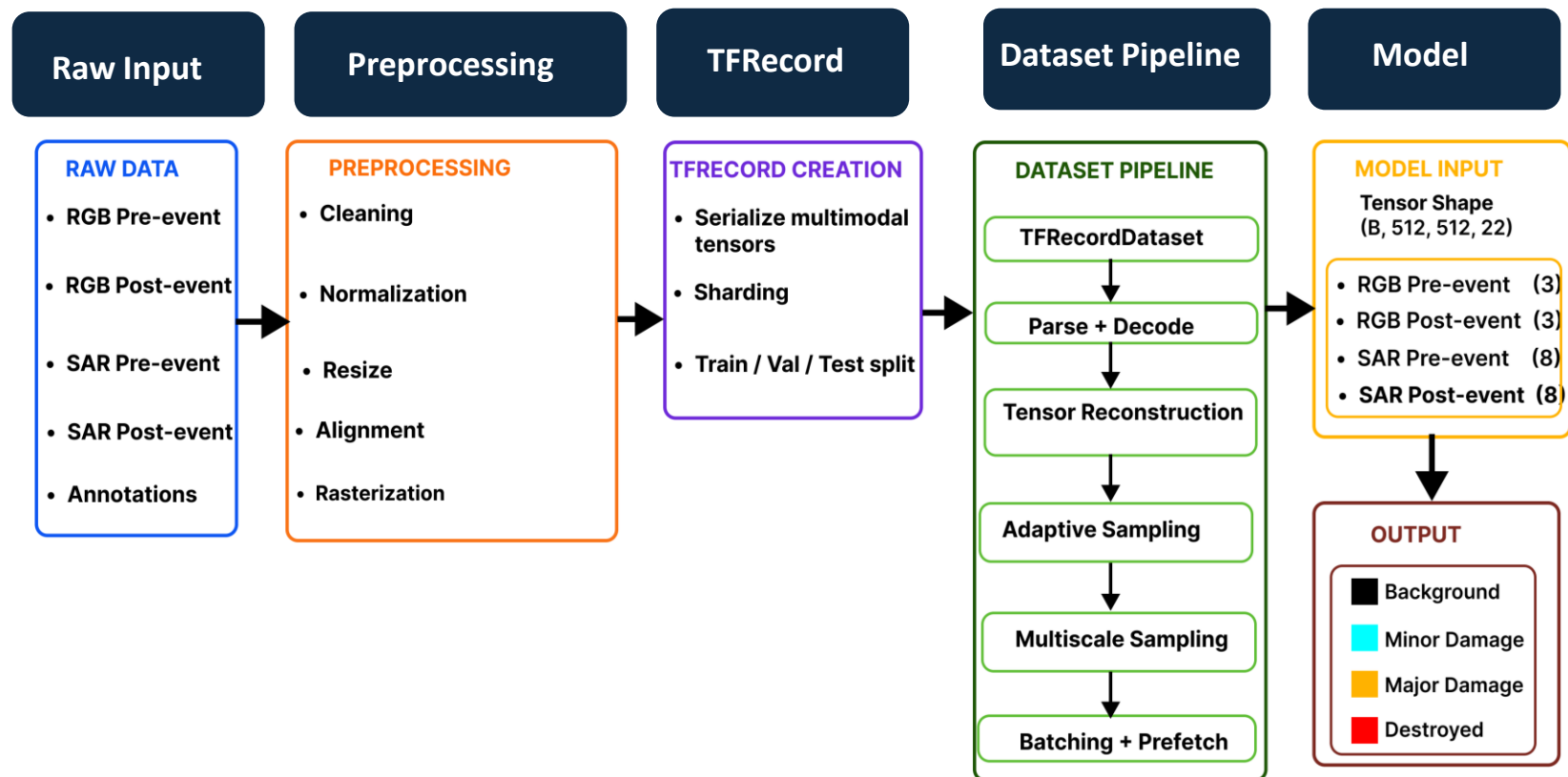
- Different resolutions (RGB vs SAR)
- Temporal misalignment
- Noise in SAR (speckle)

Processing Strategies:

RGB → ImageNet Normalization

SAR → Log scaling + clipping + light smoothing

Mask → Pixel-level labels



Efficient data pipeline enables scalable multimodal training and consistent input representation

Why Temporal Modeling is Necessary

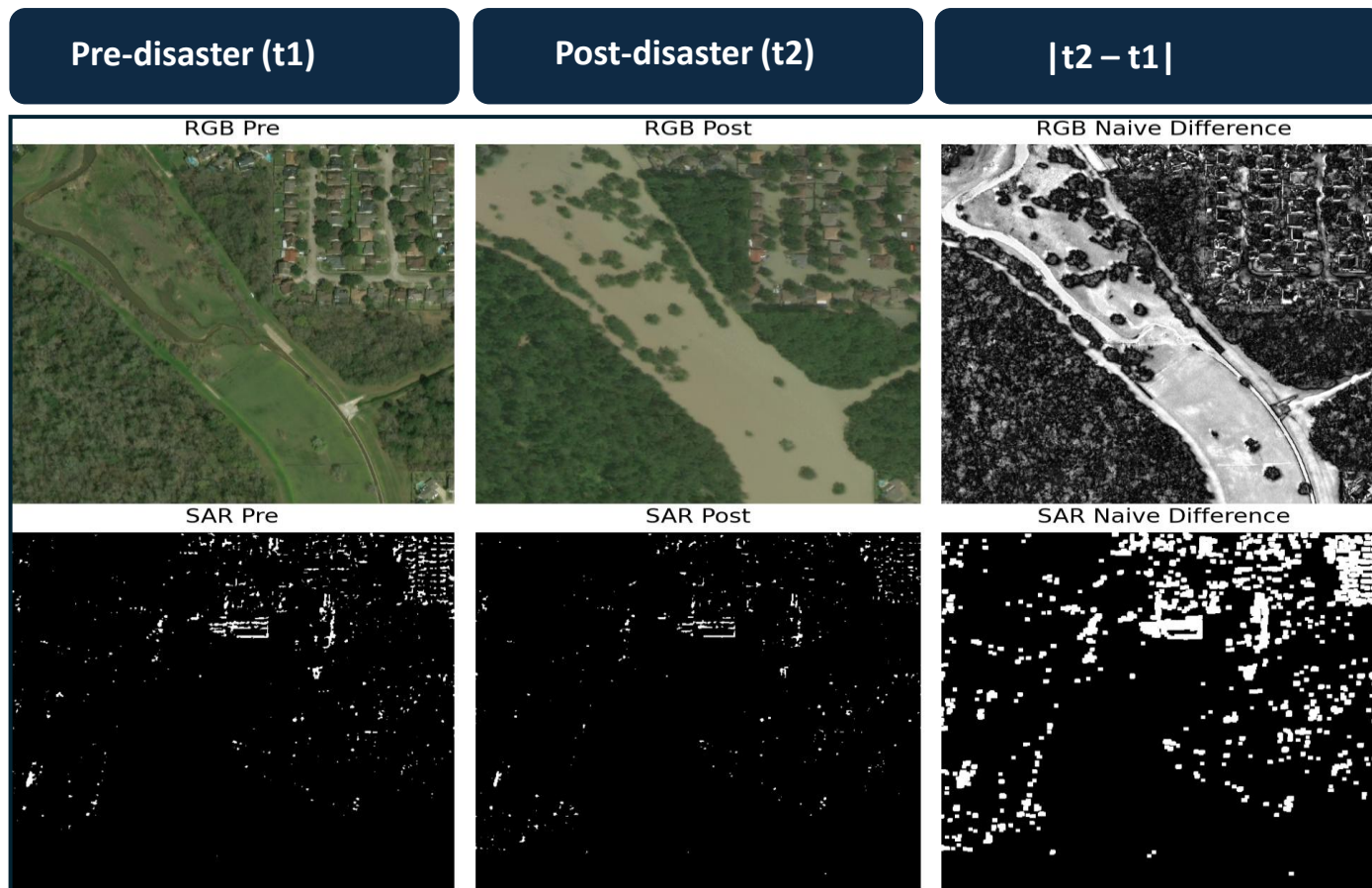
Limitations of Naïve Differencing

- Changes in RGB are subtle and easily confused with normal variations.
- Simple differencing highlights noise as much as real change.
- Handcrafted thresholds miss complex damage patterns.

Our Approach:

- We introduced a learned temporal difference (DIFF) with gating to extract meaningful change signals while suppressing noise

The temporal change signal is weak and noisy. A learning-based approach is required to extract meaningful changes while suppressing noise



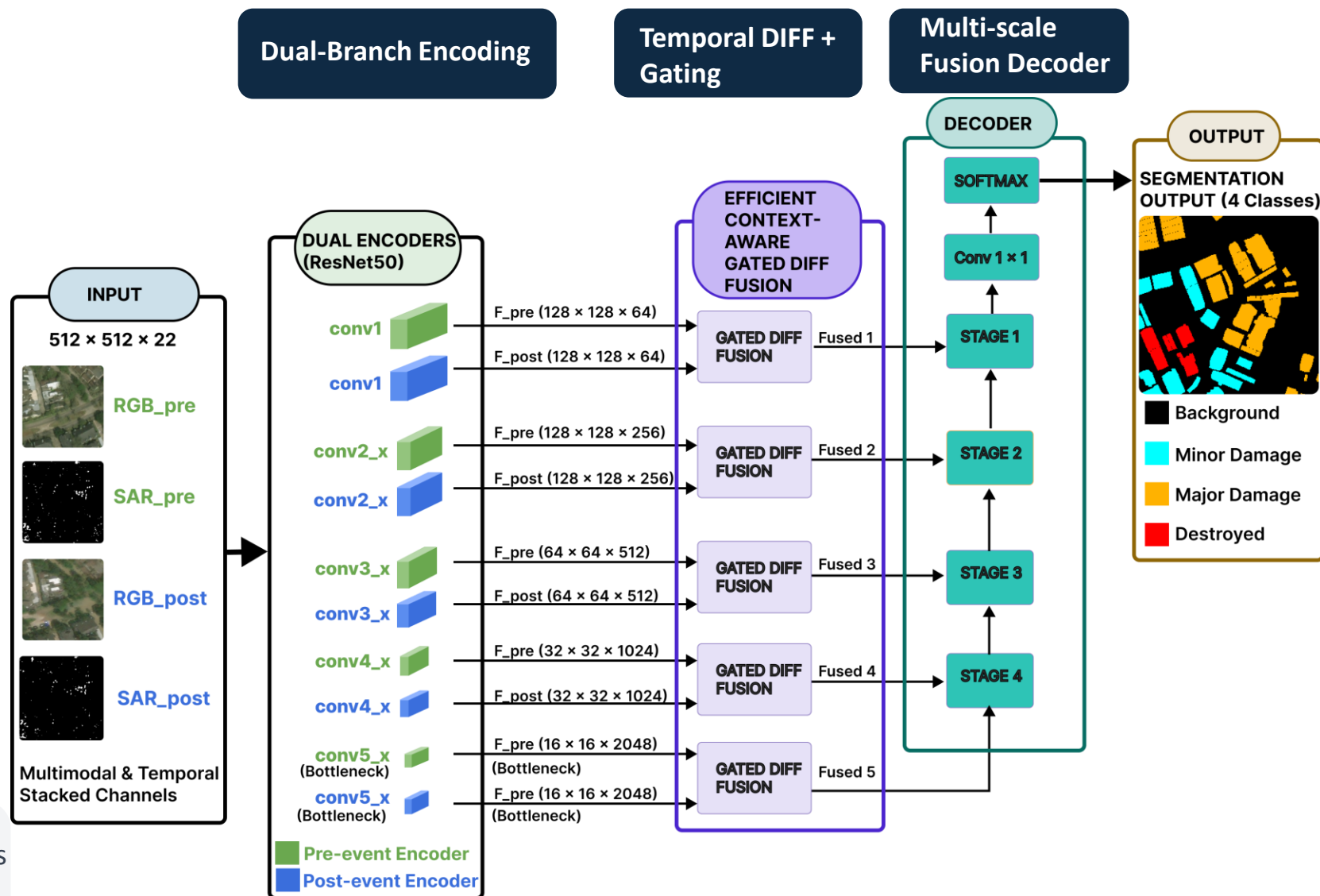
Key Challenge: Detecting structural changes between pre- and post- disaster images is difficult due to subtle visual differences, noise and ambiguities.

Model Architecture – Multimodal Temporal Fusion

Core Designs

- Dual-branch encoders (pre / post)
- Explicit learned temporal modeling via DIFF)
- Context-aware gated fusion
- Multi-scale feature aggregation (UNET decoder)

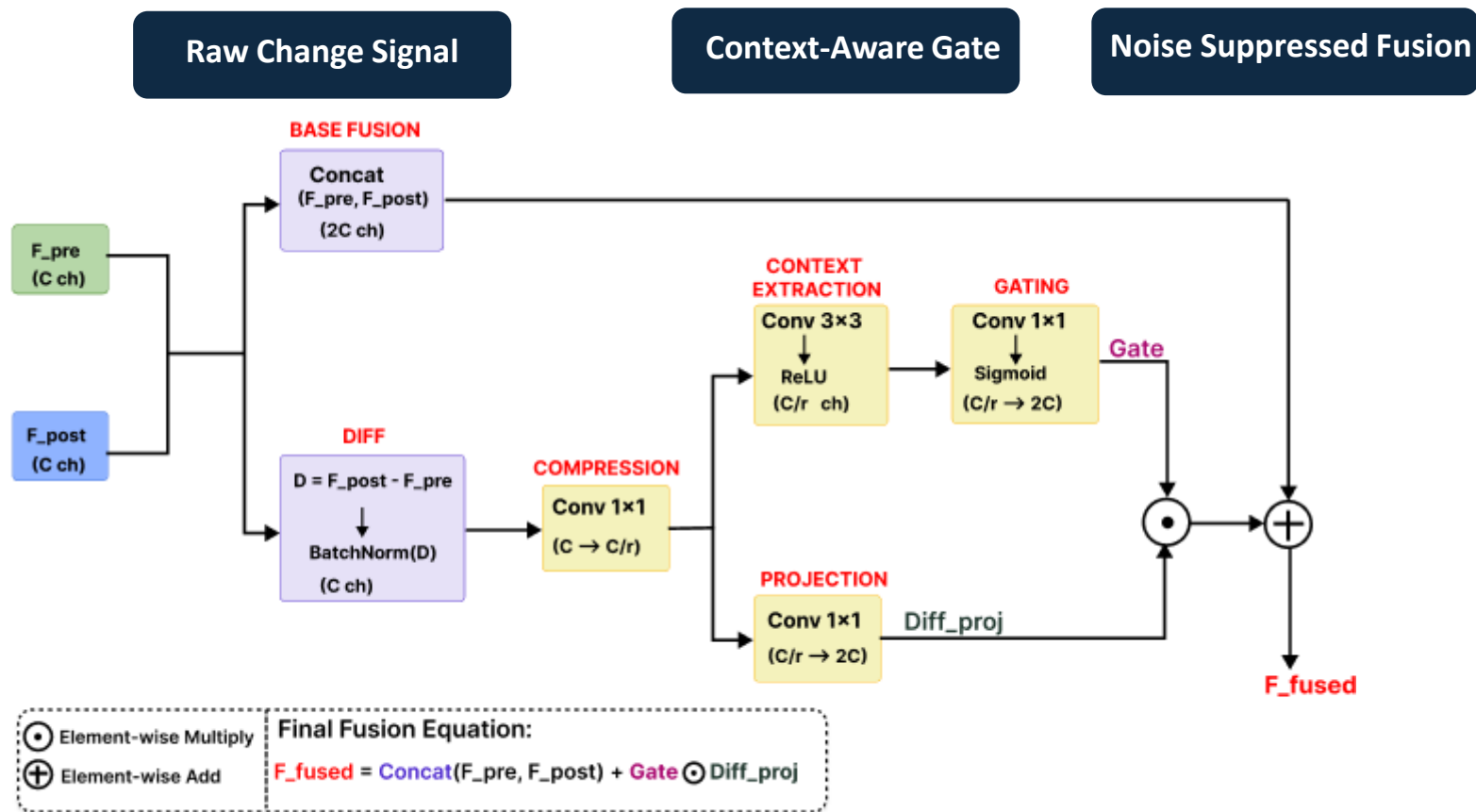
Our architecture explicitly models temporal change and fuses multimodal information at different scales to generate accurate and consistent damage predictions



Key Innovation: Efficient context-aware Gated DIFF Fusion

Mechanism: Efficient Context-Aware Gated Fusion

- **Extract** explicit temporal change between pre- and post-disaster features
- **Project** to a compact representation
- **Learn spatial gate** from context to evaluate change reliability
- **Suppress noise** by filtering the difference between the learned gate



Not all change is meaningful – we learn to filter change using context

Naïve (Diff only):

Amplifies noise and false signals

Context-aware gated DIFF:

Suppresses noise, highlights meaningful changes

Training Strategy

Balanced TFRecords

(class imbalance handling)

Damage-aware sampling

(Destroyed → Damage → Random)

Multi-scale training

(Multiscale Sampling => 50% : 512, 30% :384, 20% :256 → 512)

Combined Loss

(CCE + Dice)

Designed to stabilize training under class imbalance and noisy multimodal inputs

Final Performance

Overall Metrics

mIoU: 0.5136

Damage IoU: 0.3564

Per-class IoU

Background: 0.9887

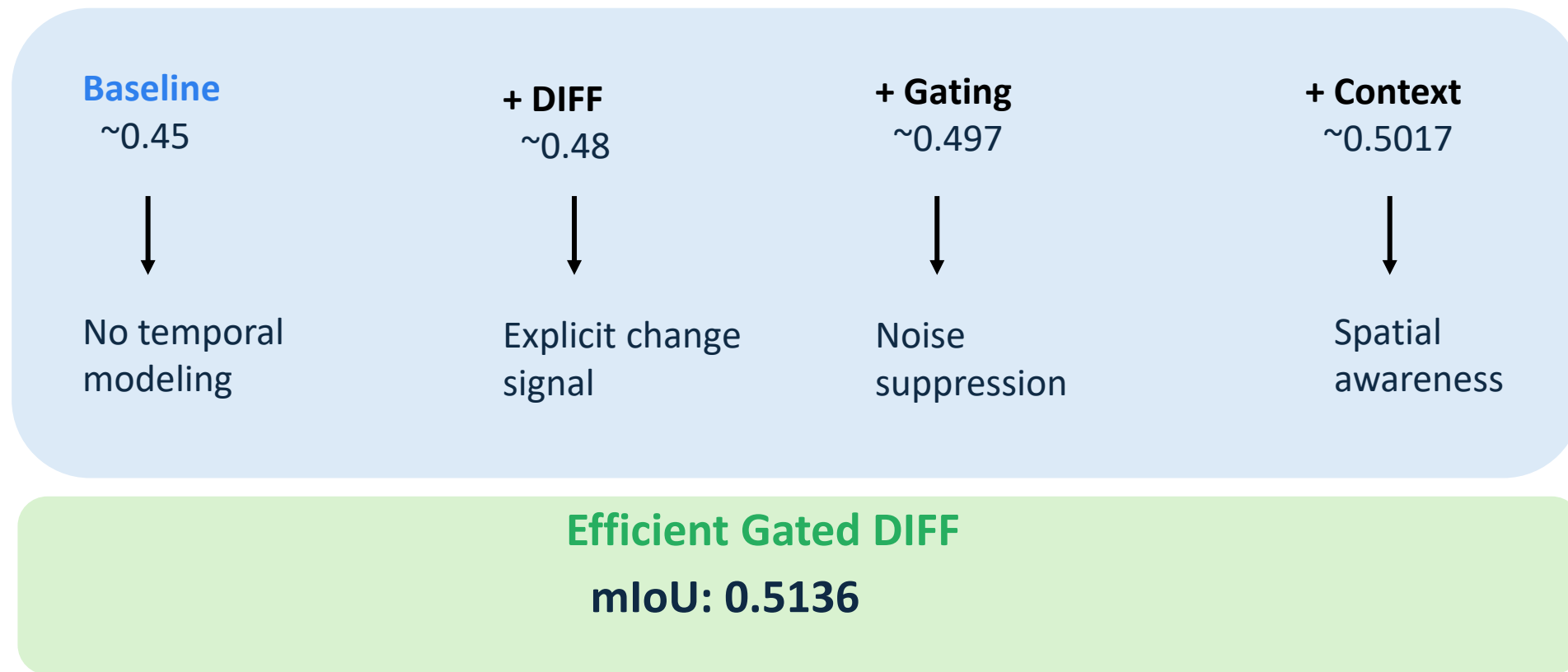
Minor: 0.1685

Major: 0.3971

Destroyed: 0.4999

Test set (unseen)

Model Evolution



Baseline: Dual-branch ResNet50 UNet (no temporal modeling)

Each step incrementally improves performance

Systems Impact – Efficient Gated DIFF (v3)

Gated DIFF (v2)

mIoU: 0.5007

Latency: ~2400 ms

Model Size: ~4.8 GB

Memory: ~4.6 GB

Efficient Gated DIFF (v3)

0.5136 (+0.013)

~34 ms (~70× faster)

~1.1 GB (~4× smaller)

~1.2 GB (~4× lower)

- ✓ Higher accuracy
- ✓ ~70× faster inference
- ✓ ~4× smaller model
- ✓ Real-time capable (~30 FPS)

Better AND faster → substantially more deployable

Critical Debugging – SAR Temporal Failure

Problem

SAR_pre = SAR_post → No temporal signal
(Model cannot learn change → behaves like single image)

Root Cause

Incorrect temporal window during SAR download

Fix

Correct pre/post temporal windows in data pipeline

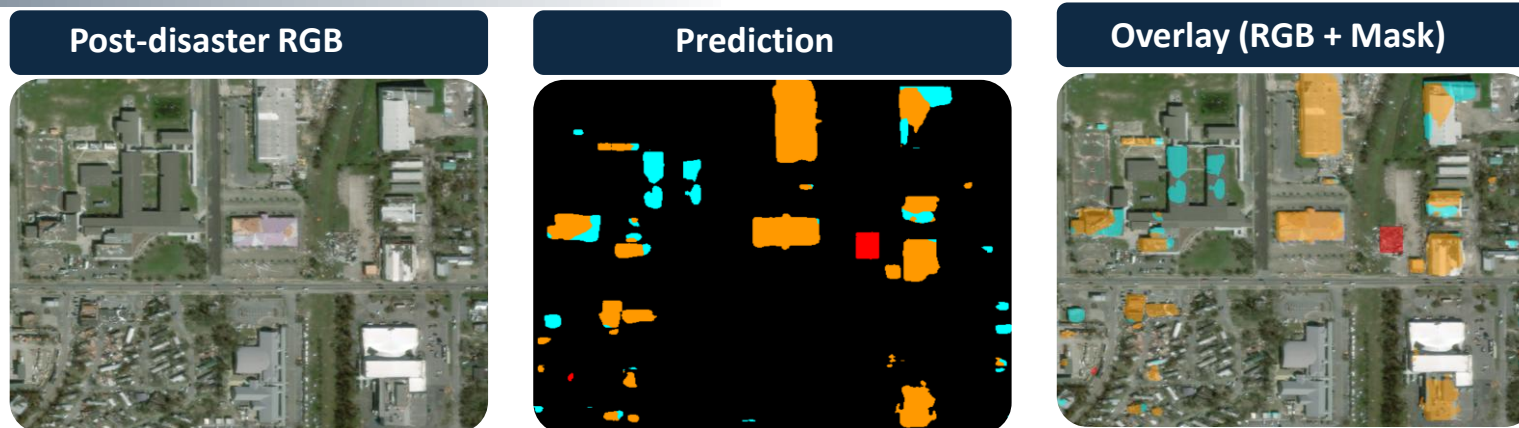
Impact

- ✓ Restored true temporal signal
- ✓ Enabled DIFF-based learning
- ✓ Significant improvement in damage detection

Data quality directly determines model performance

Visual Results – Model Behavior Across Scenarios

BEST CASE →



TYPICAL CASE →



FAILURE CASE →
(Edge scenario)



Model performs reliably but struggles in highly ambiguous scenes (e.g., occlusion, low contrast, clutter).

Conclusion

- ✓ **Temporal modeling is essential for change detection**
- ✓ **Context-aware gated fusion improves robustness**
- ✓ **Efficient design enables real-time deployment**

Multimodal + temporal learning → reliable damage detection